COMPUTER ENGINEERING

QUEZON CITY

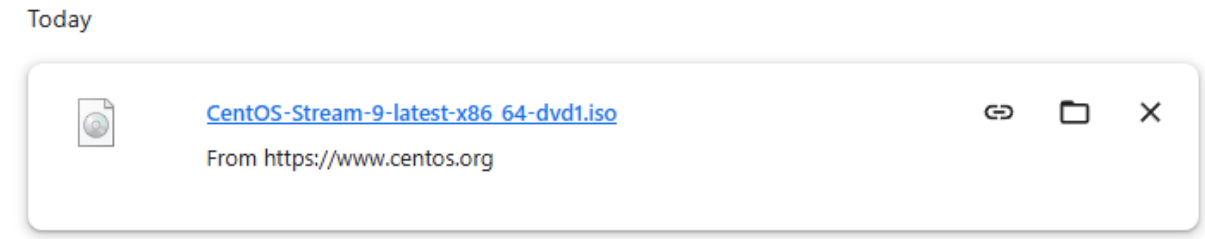
COURSE CODE/TITLE:	**CPE212 / Automating Server Management**	**SECTION:**	**CPE31S2**
DATE PERFORMED:	**Jul 23 2026**	**DATE SUBMITTED:**	**Jul 23 2026**
NAME:	**Kenzo Lim**	**INSTRUCTOR:**	**Engr. Jimlord Quejado**

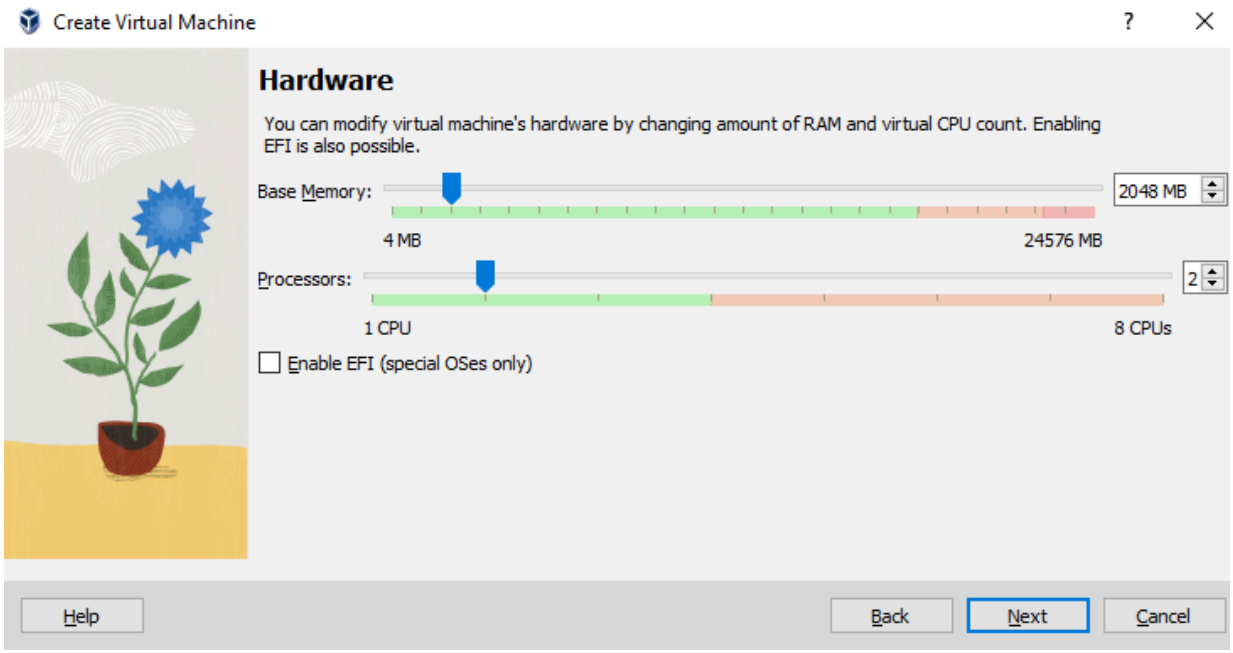
ACTIVITY NO. 3.1 Install SSH server on CentOS or RHEL 8

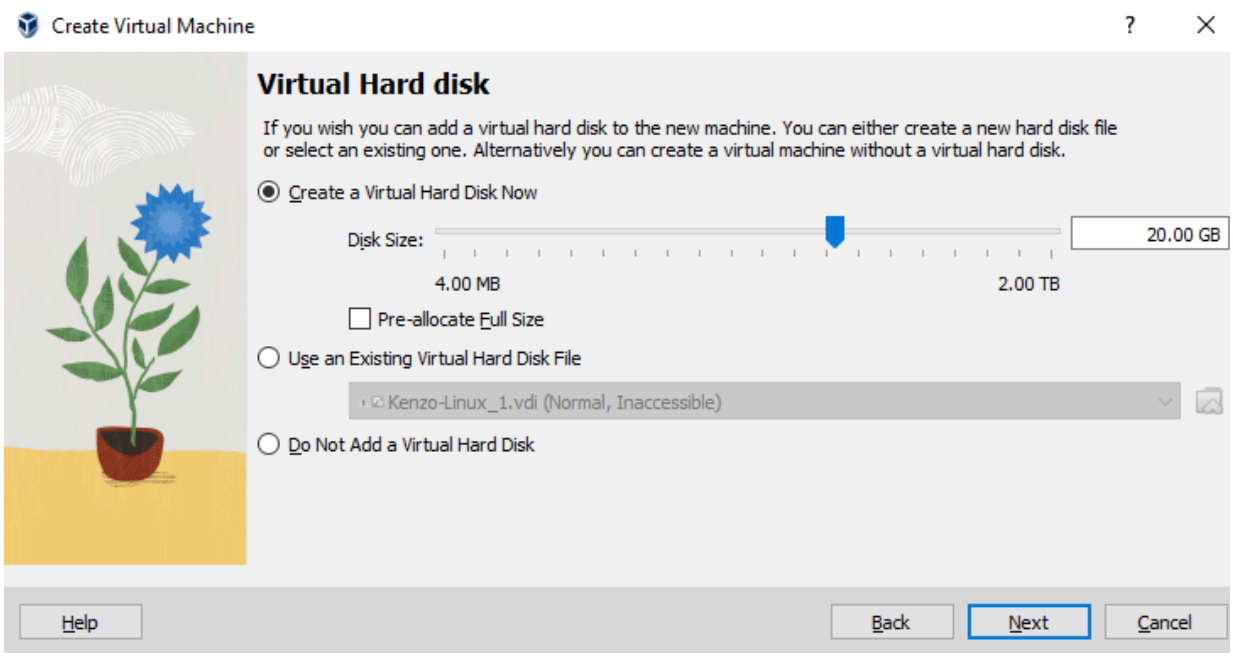
1. PROCEDURE OUTPUT

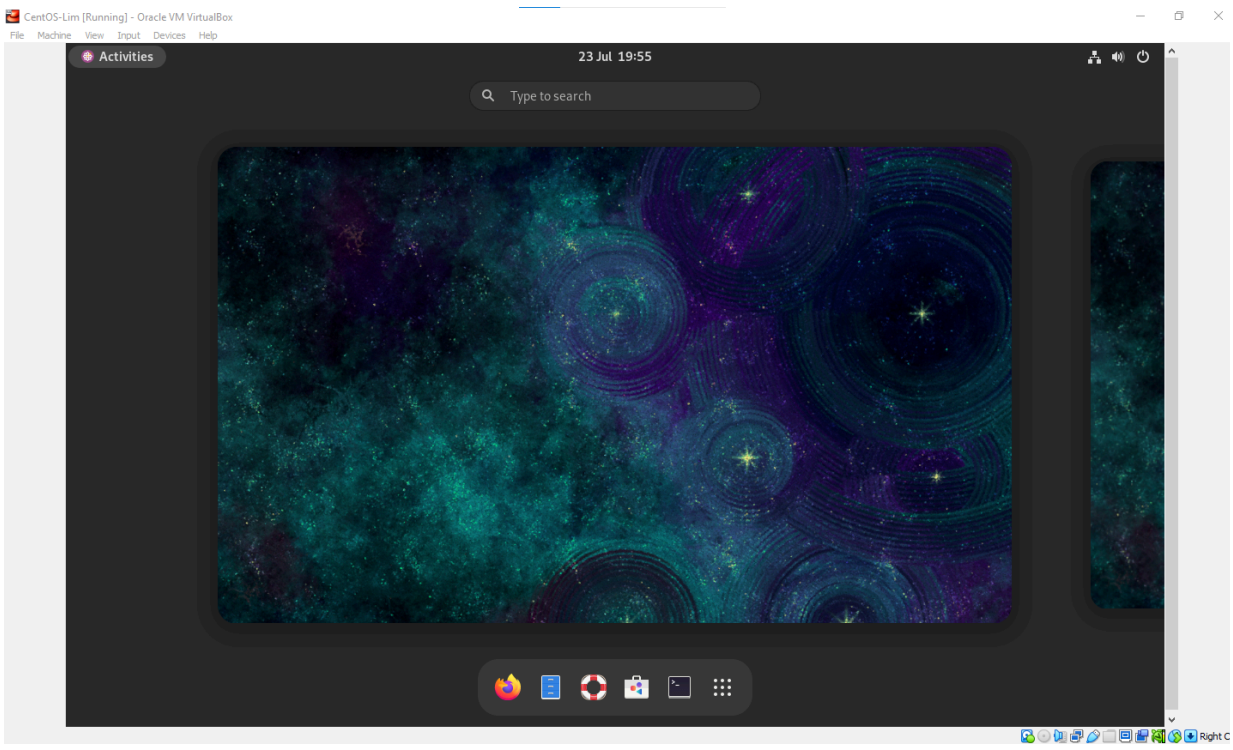
Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS here: http://mirror.rise.ph/centos/7.9.2009/isos/x86_64/
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.
3. Install the downloaded image.
4. Show evidence that the OS was installed already.



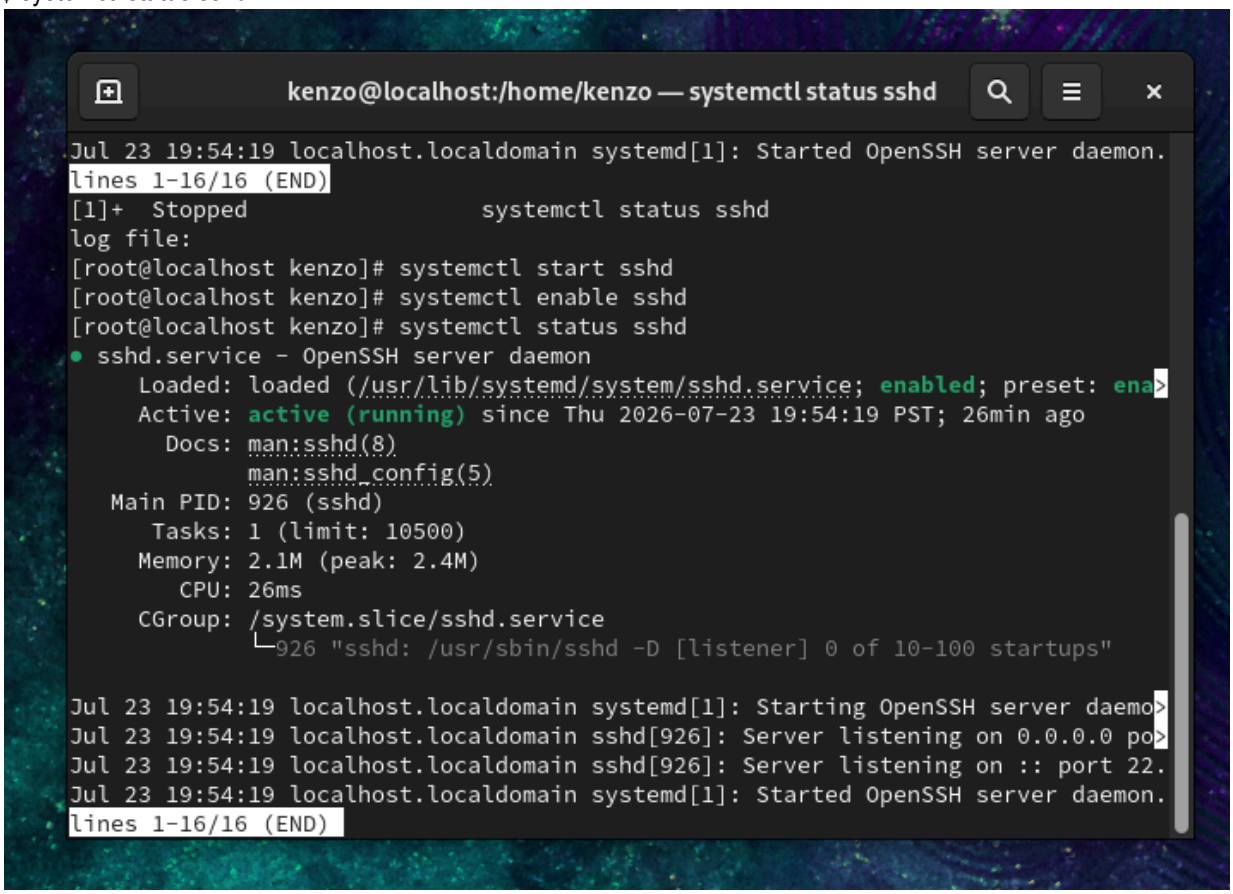




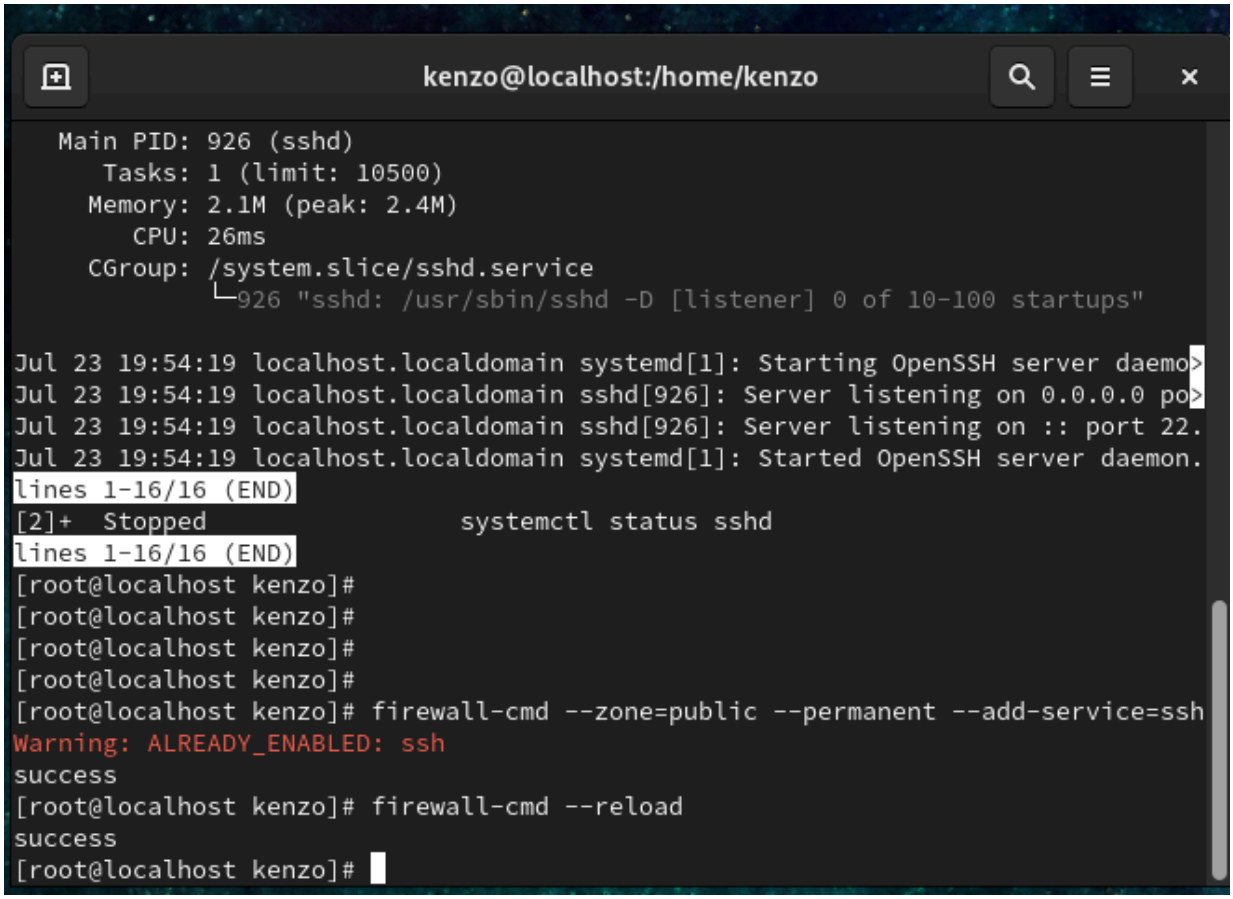


Task 2: Install the SSH server package openssh
1. Install the ssh server package openssh by using the dnf command:
$ dnf install openssh-server
2. Start the sshd daemon and set to start after reboot:
$ systemctl start sshd
$ systemctl enable sshd

3. Confirm that the sshd daemon is up and running:
$ systemctl status sshd

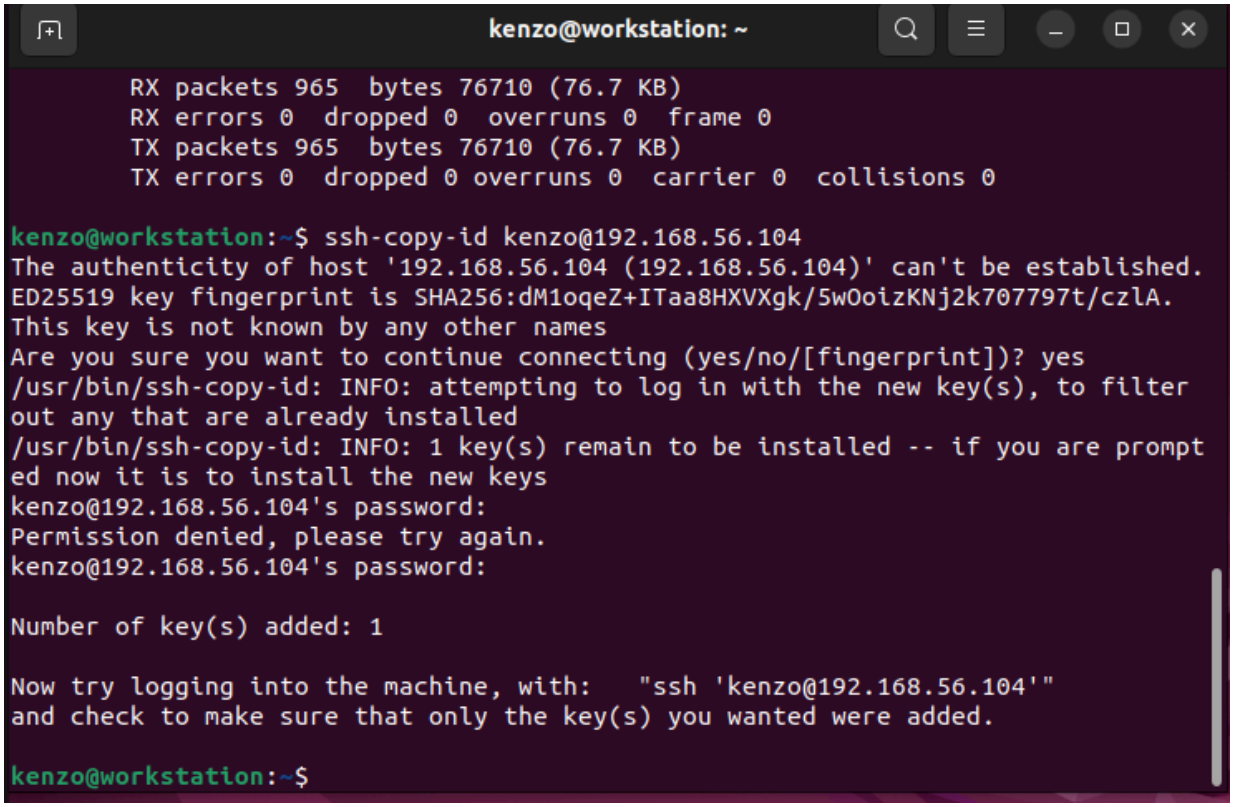


4. Open the SSH port 22 to allow incoming traffic:
$ firewall-cmd --zone=public --permanent --add-service=ssh
$ firewall-cmd --reload



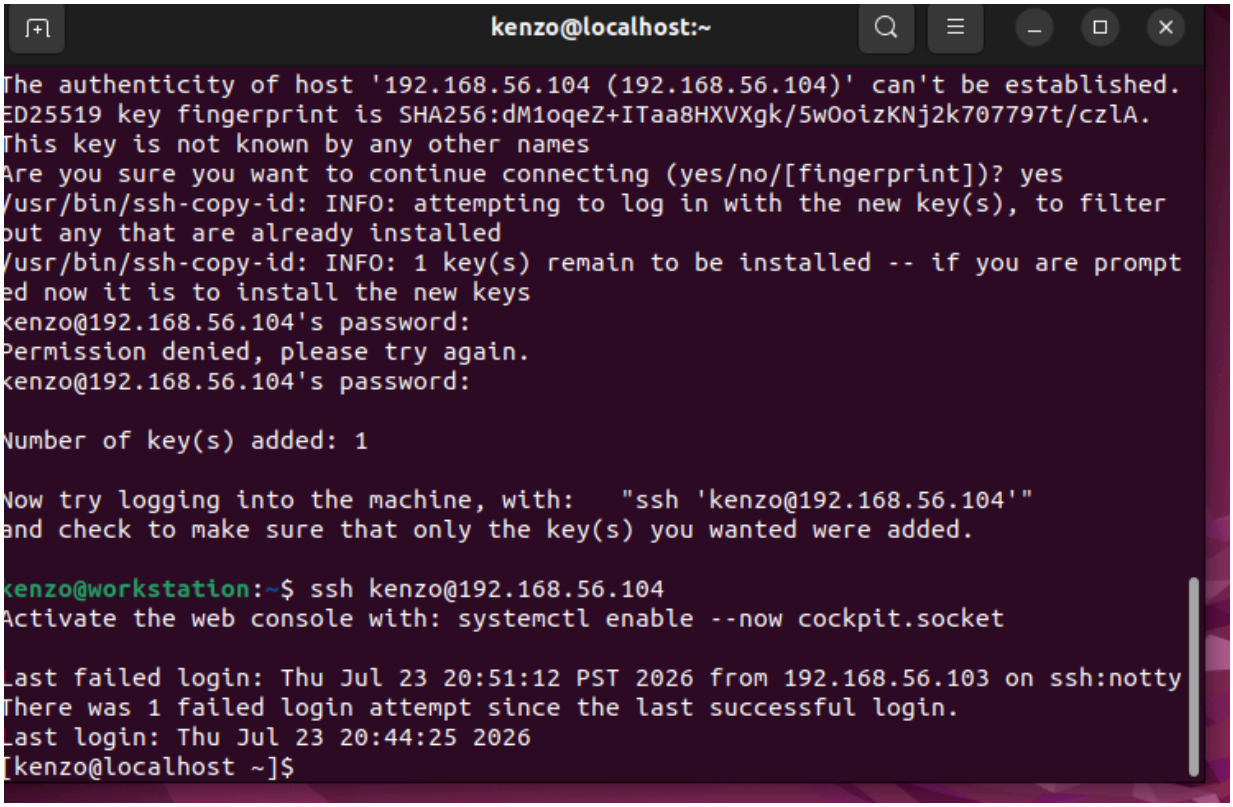
5. Locate the ssh server man config file /etc/ssh/sshd_config and perform custom configuration. Every time you make any change to the /etc/ssh/sshd-config configuration file reload the sshd service to apply changes:
$ systemctl reload sshd

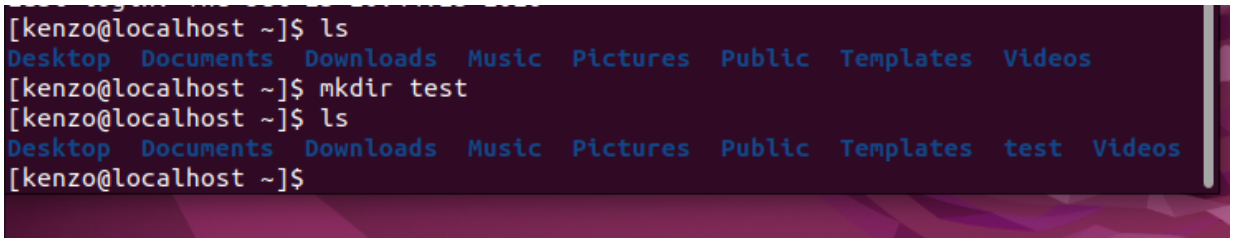
Task 3: Copy the Public Key to CentOS
1. Make sure that ssh is installed on the local machine.
2. Using the command ssh-copy-id, connect your local machine to CentOS.
3. On CentOS, verify that you have the authorized_keys.

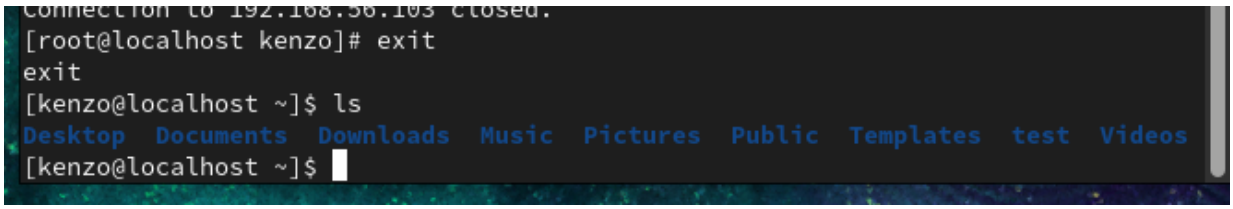


Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.
2. Show evidence that you are connected







2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?

I personally think it depends on the user's needs, although the purpose and stability, as well as reliability is a must when choosing between Debian and RHEL, both are stable, but Debian is known for its conservative updates, while Red Hat is designed for enterprise-grade reliability. Users should also consider the cost of the system, Debian is free while RHEL typically requires a paid subscription for official support.

2. What are the main differences between Debian and Red Hat Linux distributions?

The differences they have would be their developer first, Debian is a community-driven Debian project while RHEL is IBM subsidiary, second would be their cost, Debian is free and open source while RHEL is subscription-based for enterprise support. They also have different package format such as .deb for Debian, and .rpm for RHEL, the package manager are different as well, APT for Debian and DNF(Dandified Yum) which replaced the older YUM package.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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